When to Use

To compare the means of two or more quantitative variables obtained from dependent samples (repeated measures or matched groups). The two or more scores might be the same variable measured at two or more different times or under different conditions, two or more comparable variables measured at the same time, or some combination.

Research Hypothesis: Actors playing Clark Kent/Superman are older on average than actors playing Lois Lane across Superman media portrayals.

Null Hypothesis: The mean ages of actors playing Clark Kent and Lois Lane in Superman portrayals are the same.

Research Design

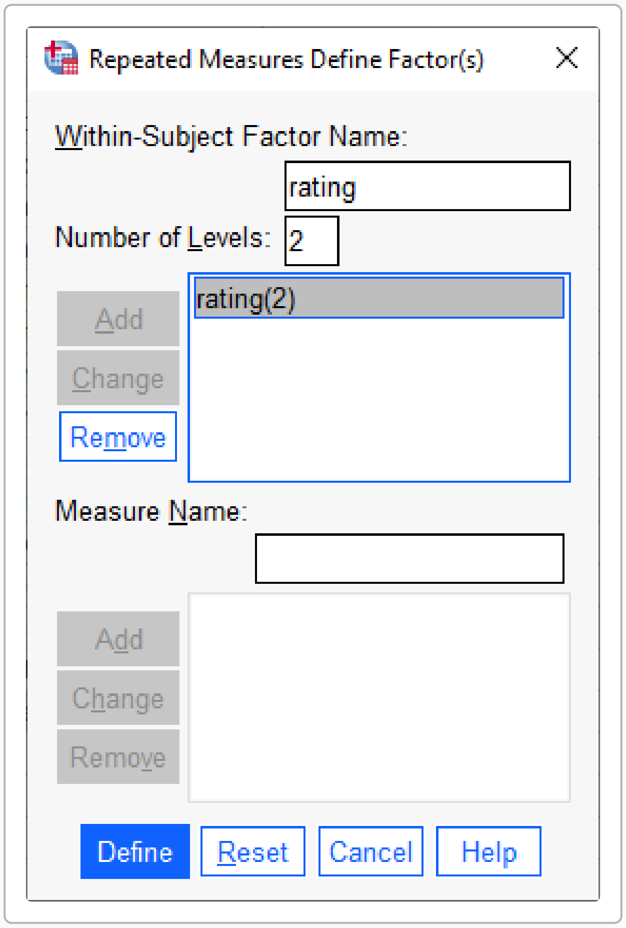
The IV is Character Role, with the conditions Clark Kent and Lois Lane. The DV is the actor's age at time of portrayal.

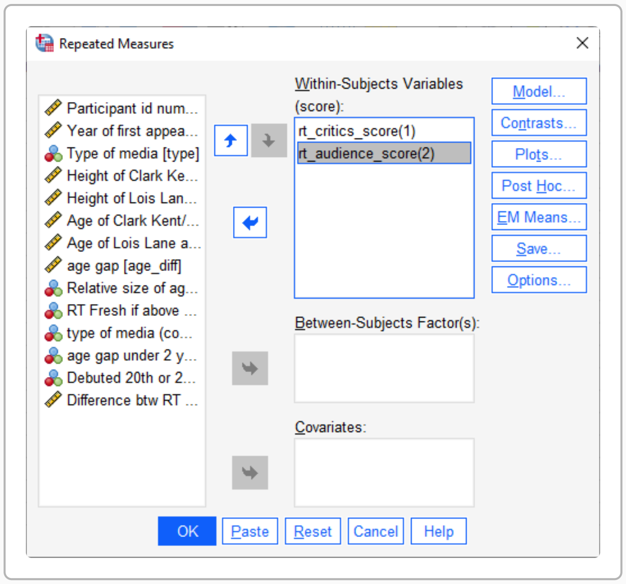
Clark Kent	Lois Lane

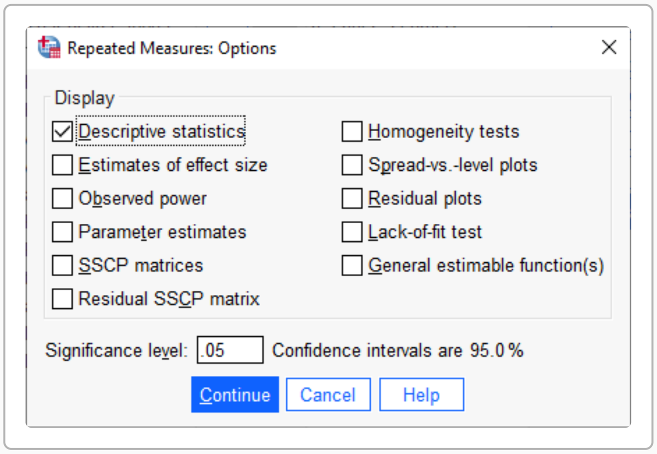
Variables in the Analysis: In a WG design the variables in the analysis are the variables holding the DV scores for each IV condition (clark_age & lois_age).

Analyze → General Linear Model → Repeated Measures

- In the **Repeated Measures Definition Window** enter your name for the IV in the “Within-subject Factor Name” box (Char-Role)
- Enter the number of conditions of the IV in the “Number of levels” window (2)
- Click the “Add” button
- Click the “Define” button
- In the **Repeated Measures** window highlight the variables that are the DV score for each condition and click the arrow
- Click the “Options” button — in the **Repeated Measures: Options** window check the “Descriptives” box







SPSS Syntax

```
GLM clark_age lois_age          ①
  /WSFACTOR=CharRole 2          ②
  /PRINT=DESCRIPTIVE.           ③
```

① List variables holding DV for each IV condition
② Give a name to the WG IV & tell # conditions
③ Get mean and std

SPSS Output

Descriptive Statistics

	Mean	Std. Deviation	N
clark_age	30.58	4.883	9
lois_age	30.17	6.408	9

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
Sig.					
CharRole	Sphericity Assumed	0.728	1	0.728	0.027
.873	Error(CharRole)	Sphericity Assumed	215.001	8	26.875

SPSS provides different "versions" of the ANOVA output. We will focus on the "traditional" analysis, which SPSS labels as "Sphericity Assumed".

- **df effect** & **F** & **p-value** to use
- **df(error)** & **Mean Square Error** term for this analysis

The p-value of 0.873 means that there is about a 87.3% chance that this result is a Type I error.

Remember, even if the printout shows p = .000, never report p = .000, because that would suggest there is no possibility of a Type 1 error. Instead, report "p < .001".

Reporting the Results

Sample APA Write-Up

Figure/Table 1 shows the mean actor ages for Clark Kent and Lois Lane across 9 Superman portrayals. Clark Kent (*M* = 30.58, *SD* = 4.88) and Lois Lane (*M* = 30.17, *SD* = 6.41). There was no significant difference between the mean actor ages, $F(1,8) = 0.03$, $MSe = 26.88$, $p = 0.873$. Contrary to the research hypothesis, there was no significant difference between the mean ages of actors playing Clark Kent and Lois Lane.

It is important to report the univariate statistics for the dependent variable for both groups before presenting the ANOVA results. Often these are presented in a table or a figure.

As in the example, be sure to communicate:

- The research hypothesis (if there is one)
- The statistical results
- Whether or not those results support the research hypothesis

Sometimes, the univariate statistics are presented in text, along with the ANOVA results.